The due date for submitting this assignment has passed.
Due on 2026-05-08, 23:59 IST.

You may submit any number of times before the due date. The final submission will be considered for grading.

1) Choose all the subspaces isomorphic to R^4. ***2 points***

- [ ] $W_1 = \left\{ \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} : a+b+c+d+e+f = 0 \right\}$
- [ ] $W_2 = \{(x,y,z,w,u) : x+y+z = w,\ u = 0\}.$
- [ ] $W_3 = \{ \begin{bmatrix} a & b & c & d \\ e & f & g & h \end{bmatrix} : a+b+c+d = 0,\ e+f+g+h = 0\}.$
- [ ] $W_4$is the column space of the matrix $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.
- [ ] W_5 is the orthogonal complement of the zero subspace in R^4 with respect to the standard inner product. \end{enumerate}

No, the answer is incorrect.

Score: 0

Accepted Answers:

$W_4$is the column space of the matrix $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

W_5 is the orthogonal complement of the zero subspace in R^4 with respect to the standard inner product. \end{enumerate}

2) Consider a basis $\{(1,0,1),\ (-1,1,1)\}$ of a subspace W in R^3 with the standard dot product. Find the projection vector of $(1,2,3)$ onto W. ***2 points***

- ( ) $(\frac{2}{3}, \frac{4}{3}, \frac{10}{3})$
- ( ) $(-\frac{2}{3}, -\frac{4}{3}, -\frac{10}{3})$
- ( ) $(-\frac{2}{3}, \frac{4}{3}, -\frac{10}{3})$
- ( ) $(\frac{2}{3}, -\frac{4}{3}, \frac{10}{3})$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(\frac{2}{3}, \frac{4}{3}, \frac{10}{3})$

Consider the vector space R^3 with usual addition and scalar multiplication. Let S be its subset defined as follows:

$$S = \{(1, 0, 1), (0, 1, 1)\}$$

and W_1 and W_2 be its vector subspaces defined as:

$$W_1 = \{(x, y, z) \mid x = z, \text{ and } y = z; x, y, z \in \mathrm{R}\}$$
$$W_2 = \{(x, y, z) \mid z = x + y; x, y, z \in \mathrm{R}\}$$

Answer the following questions:

3) Which of the following options is (are) true? ***2 points***

- [ ] S is linearly independent.
- [ ] S spans the vector space R^3 (with usual addition and scalar multiplication).
- [ ] $Span(S)$ is a proper subspace of R^3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

S is linearly independent.
$Span(S)$ is a proper subspace of R^3.

4) Which of the following options is (are) correct? ***1 point***

- ( ) S is a basis of W_1.
- ( ) S is a basis of W_2.

No, the answer is incorrect.

Score: 0

Accepted Answers:

S is a basis of W_2.

5) What is the dimension of W_1?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

6) What is the dimension of W_2?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Consider a map $T : V \rightarrow W$, defined by $T(A) = trace(A)$, where V is the vector space of all matrices of order 2 with respect to the usual addition and scalar multiplication in matrices, and W is R with usual addition and scalar multiplication. Let $\beta = \{\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 1\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\1 & 0\end{bmatrix} \begin{bmatrix}0 & 0\\0 & 1\end{bmatrix}\}$ be an ordered basis of V and $\gamma = \{1\}$ be an ordered basis of W. Answer the following questions using the information above.

7) Choose the set of correct options from the following: ***2 points***

- [ ] $T(A+B) = T(A) + T(B)$ for all $A, B \in V$.
- [ ] $T(cA) = cT(A)$ for all $A \in V$, and $c \in \mathrm{R}$.
- [ ] T is not a linear transformmation.
- [ ] $T(I) = 0$, where I denotes the identity matrix of order 2.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T(A+B) = T(A) + T(B)$ for all $A, B \in V$.
$T(cA) = cT(A)$ for all $A \in V$, and $c \in \mathrm{R}$.

8) Choose the set of correct options from the following: ***2 points***

- ○ If T is a linear transformation and M is the matrix representation of T with respect to the bases β for the domain and γ for the codomain, then $M = [1 \quad 0 \quad 0 \quad 1]$.
- ○ If T is a linear transformation and M is the matrix representation of T with respect to the bases β for the domain and γ for the codomain, then $M = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$.
- ○ If T is a linear transformation and M is the matrix representation of T with respect to the bases β for the domain and γ for the codomain, then $M = [1 \quad 0 \quad 1 \quad 0]$.
- ○ If T is a linear transformation and M is the matrix representation of T with respect to the bases β for the domain and γ for the codomain, then $M = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If T is a linear transformation and M is the matrix representation of T with respect to the bases β for the domain and γ for the codomain, then $M = [1 \quad 0 \quad 0 \quad 1]$.

9) Let A and B be 4×4 real matrices such that $\text{rank}(A) = 2$ and $\text{rank}(B) = 3$. ***3 points***
Choose all the correct options from the following:

- [ ] A and B are not equivalent.
- [ ] The system $Ax = 0$ has infinitely many solutions and the solution space has dimension 2.
- [ ] The nullity of B is 1, and hence $Bx = 0$ has a unique non-trivial solution up to scalar multiples.
- [ ] The columns of A lie in a subspace of R^4 of dimension 2, while the columns of B span R^4.
- [ ] If u_1, u_2, u_3, u_4 are the columns of B, then these vectors must be linearly dependent but cannot all lie in the same plane.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A and B are not equivalent.
The system $Ax = 0$ has infinitely many solutions and the solution space has dimension 2.
The nullity of B is 1, and hence $Bx = 0$ has a unique non-trivial solution up to scalar multiples.
If u_1, u_2, u_3, u_4 are the columns of B, then these vectors must be linearly dependent but cannot all lie in the same plane.

10) Choose the correct statements for the function $f(x, y) = \begin{cases} \frac{x^2y^2}{x^4+y^4} & (x,y) = (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$ ***3 points***

- [ ] $\lim_{(x,y)\to(0,0)} f(x,y) = 0$
- [ ] f is not continuous at $(0, 0)$.
- [ ] $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist.
- [ ] $\lim_{(x,y)\to(1,-1)} f(x,y) = \frac{1}{2}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

f is not continuous at $(0, 0)$.

$\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist.

$\lim_{(x,y)\to(1,-1)} f(x, y) = \frac{1}{2}$

11) ***2 points***

From the options below, choose all the options containing functions defined on the domain $\mathbb{R}^2 \setminus \{(0, 0)\}$ for which the limit of the function at the origin along any straight line through the origin exists, and the limits are all equal.

- ☐ $f(x, y) = \frac{(x-y)^2}{x^2+y^2}$
- ☐ $f(x, y) = \frac{xy}{\sqrt{x^2+y^2}}$
- ☐ $f(x, y) = (x^2 + y^2) \sin\left(\frac{1}{x^2+y^2}\right)$
- ☐ $f(x, y) = \frac{xy^2}{x^2+y^4}$
- ☐ $f(x, y) = \frac{x^2-y^2}{x^2+y^2}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$f(x, y) = \frac{xy}{\sqrt{x^2+y^2}}$$

$$f(x, y) = (x^2 + y^2) \sin\left(\frac{1}{x^2+y^2}\right)$$

$$f(x,y) = \frac{xy^c}{x^2+y^4}$$

12) Consider the following functions f defined as: ***2 points***

$$f(x,y,z) = xyz.$$

Then the tangent line at the point (1,0,1) in the direction of (1,1,1) is parallel to the line

- $x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}$
- $x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = t\sqrt{3}$
- $x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = 1 + \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}$
- None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}$

Let $f : \mathrm{R}^2 \rightarrow \mathrm{R}$ be the real-valued function defined by $f(x, y) = x^2 - xy + y^2$, for all $(x, y) \in \mathrm{R}^2$. Let $T(3, 1)$ denote the tangent plane to the graph of f at the point $(3, 1)$. Since $T(3, 1)$ is an affine subspace of R^3, there exists a vector $v \in \mathrm{R}^3$ and a unique subspace $U \subseteq \mathrm{R}^3$ such that $T(3, 1) = v + U = \{v + u \mid u \epsilon U\}$.

13) Which of the following are possible candidates for v ***2 points***

- [ ] $(2, 1, 2)$
- [ ] $(0, 7, 0)$
- [ ] $(1, -2, 0)$
- [ ] $(2, 0, -1)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(2, 1, 2)$
$(1, -2, 0)$

14) Which of the following is a basis for U? ***2 points***

- ( ) $\{(1, -1, -1), (1, 3, 2)\}$
- ( ) $\{(1, 2, 3), (-1, -1, -1)\}$
- ( ) $\{(0, 1, -1), (0, -1, 1)\}$
- ( ) $\{(1, 2, 3), (0, -1, 1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1, 2, 3), (0, -1, 1)\}$

Consider the function $f(x, y) = x^3 - 3x + y^3 - 3y^2$.

15) Which of the following points are the critical points of f? ***2 points***

- [ ] (1,0)
- [ ] (0,0)
- [ ] (-1,0)
- [ ] (0,1)
- [ ] (1,2)
- [ ] (2,1)
- [ ] (-1,2)
- [ ] (2,-1)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(1,0)
(-1,0)
(1,2)
(-1,2)

16) Choose the correct options from the following. ***2 points***

- [ ] (1,0) is a local minima of the function f.
- [ ] (0,0) is a local minima of the function f.
- [ ] (1,2) is a local minima of the function f.
- [ ] (-1,2) is a saddle point of the function f.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(1,2) is a local minima of the function f.
(-1,2) is a saddle point of the function f.

17) Choose the correct options from the following. ***2 points***

- [ ] (-1,0) is a local maxima of the function f.
- [ ] (0,1) is a local maxima of the function f.
- [ ] (1,0) is a saddle point of the function f.
- [ ] (2,1) is a local maxima of the function f.
- [ ] (2,-1) is a local maxima of the function f.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(-1,0) is a local maxima of the function f.
(1,0) is a saddle point of the function f.

Let A_1 and A_2 be two affine spaces defined by the solution space of the system of linear equations

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$$

and

$$\begin{bmatrix} 1 & 0 & 1 \\ 2 & 0 & 2 \\ 3 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

respectively. Let V_1 and V_2 be the vector subspaces of R^3 corresponding to A_1 and A_2. Answer the following questions using the information given above.

18) Choose the correct options. ***3 points***

- V_1 has dimension 1, where as the dimension of V_2 is 2.
- V_1 has dimension 2, where as the dimension of V_2 is 1.
- Both the V_1 and V_2 have dimension 3.
- Both the V_1 and V_2 have dimension 1.

No, the answer is incorrect.

Score: 0

Accepted Answers:

V_1 has dimension 1, where as the dimension of V_2 is 2.

19) Which of the following function can represent a map from A_2 to A_1 correspond to ***3 points*** the linear transformation between their corresponding subspaces whose matrix representation is given by $[-1 \quad 0]$ with respect to the ordered bases $\{(1, 0, -1), (0, 1, 0)\}$ and $\{(1, 1, -1)\}$ for the domain and the codomain?

- $f(2 + x, y, -x) = (2 - x, 2 + x, x).$
- $f(2 + x, y, -x) = (2 + x, 2 + x, x).$
- $f(2 + x, y, -x) = (2 - x, 2 - x, x).$
- $f(2 + x, y, -x) = (2 - x, 2 - x, -x).$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f(2 + x, y, -x) = (2 - x, 2 - x, x).$

Suppose the temperature (T) at a point (x, y, z) in R^3 is given by the function

$$T(x, y, z) = e^{x+y+z}$$

Let T_1, T_2, and T_3 be the tangent planes to the function $f(x, y) = \sqrt{9 - x^2 - y^2}$ at the points $(1, 2)$, $(2, 1)$, and $(2, 2)$, respectively. The set of points on each plane T_1, T_2, and T_3 form affine subspaces A_1, A_2, and A_3, respectively, of the vector space R^3, with respect to usual addition and scalar multiplication. Let V_1, V_2, and V_3 denote the vector subspaces corresponding to the affine subspaces A_1, A_2, and A_3, respectively.

Answer the following questions using the above information.

20) Which of the following denotes the vector subspace $V_1 \cap V_2$? ***3 points***

- $\{(x, y, -x - y) \mid x, y \in \mathrm{R}\}$
- $\{(-2y - 2z, y, z) \mid y, z \in \mathrm{R}\}$
- $\{(-2z, -2z, 3z) \mid z \in \mathrm{R}\}$
- $\{(-2z, -z, z) \mid z \in \mathrm{R}\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(-2z, -2z, 3z) \mid z \in \mathrm{R}\}$

21) If m is the dimension of V_1, n is the dimension of V_2 and r is the dimension of $V_1 \cap V_2$, then find the value of $m + n + r$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 5

2 points

22) Which of the following denotes the maximum rate of change of temperature (T) at the point of intersection of the tangent planes T_1, T_2, and T_3? ***3 points***

- $\sqrt{3}e^{3}$
- $\sqrt{3}e^{\frac{27}{5}}$
- $\sqrt{3}e^{\frac{3}{5}}$
- $3e^{\frac{27}{5}}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\sqrt{3}e^{\frac{27}{5}}$

23) Which of the following options is/are true? ***2 points***

- $T(x, y, z)$ is a linear function.
- $T(x, y, z)$ is continuous function over R^3.
- $T_x(x, y, z)$ is not continuous function over R^3.
- $T(cx, cy, cz) = (T(x, y, z))^c$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T(x, y, z)$ is continuous function over R^3.
$T(cx, cy, cz) = (T(x, y, z))^c$

24) Suppose on approaching to the point $(1, 2, 3)$, temperature is approaching to e^a. What is the value of a?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 6

2 points

25) Let A be an $n \times n$ matrix such that $A^2 = 0$. Choose the correct statement from the following. ***1 point***

- Column space of A is a subset of the null space of A.
- Null space of A is a subset of the column space of A.
- Column space of A is equal to the null space of A.
- The given data is insufficient to conclude the relationship between the column space of A and null space of A.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Column space of A is a subset of the null space of A.